

Part VI — Heat, Temperature, and Thermodynamics · Chapter 30

Heat Transfer

The three mechanisms by which thermal energy moves through and between physical systems

On a winter day, touching a metal railing feels far colder than touching a wooden fence at the same outdoor temperature. Both objects are at the same temperature, yet they feel different because they conduct heat away from your hand at very different rates. The ocean moderates coastal climates because water convects heat enormous distances with great efficiency. The Sun heats the Earth across 150 million kilometers of empty space through radiation alone, with no medium of any kind. Heat transfer is not a single phenomenon — it is three distinct physical mechanisms: conduction, convection, and radiation. All three operate simultaneously in most real situations, and all three are explained by the same underlying principle: thermal energy flows from regions of higher temperature to regions of lower temperature. This chapter explores how each mechanism works, where each dominates, and how engineers and scientists exploit or combat each one.

30.1 What Heat Transfer Is

Heat transfer is the movement of thermal energy from one system to another, or from one part of a system to another, driven by a temperature difference. Heat always flows spontaneously from higher temperature to lower temperature — never in the reverse direction without external work input. This directionality is a statement of the Second Law of Thermodynamics.

Three mechanisms transfer heat: conduction (energy transfer through direct molecular contact and vibration, without bulk movement of material), convection (energy transport by the bulk motion of a fluid), and radiation (energy transfer by electromagnetic waves, requiring no material medium). These mechanisms differ fundamentally in their physical basis, their dependence on material properties, and the environments in which they operate.

In most real systems, all three mechanisms operate simultaneously. A hot radiator in a room heats nearby surfaces by radiation, heats the air in contact with it by conduction, and that warm air then rises and circulates by convection. Separating the mechanisms

analytically, as this chapter does, is a simplification that is essential for understanding each one before combining them.

30.2 Heat vs. Thermal Energy Revisited

Chapter 29 established that heat (Q) is thermal energy in transit — not a property stored in a substance, but a process of energy transfer. This chapter is entirely about the mechanisms of that transfer process.

The distinction matters here. When we say “heat flows by conduction through a wall,” we mean that thermal energy is being transferred from the warm side to the cool side by molecular interactions, without any material crossing the boundary. The wall gains internal energy on its cool side and loses internal energy from its warm side. The energy transferred is heat Q ; after transfer, it becomes part of the internal energy of the recipient.

Rate of heat transfer (power) is often more useful than the total energy transferred. The power of heat transfer $P_Q = dQ/dt$ is measured in watts ($W = J/s$). Conduction, convection, and radiation each have their own rate equations relating P_Q to the temperature difference and the geometric and material properties of the system.

30.3 Thermal Equilibrium and Heat Flow

Heat flows until thermal equilibrium is reached: the point at which all parts of a connected system are at the same temperature and no net heat transfer occurs. The approach to equilibrium is not instantaneous; it occurs over a timescale determined by the rate of heat transfer and the heat capacity of the objects involved.

The driving force for heat transfer is the temperature difference ΔT . A larger temperature difference produces a faster rate of heat transfer. When $\Delta T = 0$ (thermal equilibrium), heat transfer stops. This dependence on ΔT appears explicitly in all three transfer mechanisms: the rate of conduction is proportional to the temperature gradient; the rate of convection increases with ΔT between the surface and the fluid; the net rate of radiation is proportional to the difference of fourth powers of absolute temperatures.

This means that as two objects exchange heat and approach equilibrium, the rate of transfer automatically slows — not because any physical property of the materials changes, but because ΔT decreases. The approach to equilibrium is exponential: the temperature difference decreases by a constant fraction per unit time (Newton’s Law of

Cooling for convective/conductive situations).

30.4 Conduction

Conduction is the transfer of thermal energy through a material without bulk movement of the material itself. Energy is transferred by molecular collisions and vibrations: energetic molecules in the hot region collide with and transfer energy to adjacent less-energetic molecules, which transfer energy to their neighbors, and so on through the material.

In metals, conduction is enhanced by free electrons that carry kinetic energy rapidly from hot to cool regions. This is why metals are excellent thermal conductors and why good electrical conductors are generally also good thermal conductors. In insulators and non-metallic solids, only lattice vibrations (phonons) carry heat, which is much slower.

The rate of heat conduction through a slab of material is given by Fourier's Law of Heat Conduction:

$$P = Q/t = kA\Delta T/d$$

Fourier's Law: conduction rate P (watts). k = thermal conductivity ($W/m\cdot K$); A = cross-sectional area (m^2); ΔT = temperature difference (K or $^{\circ}C$); d = thickness (m).

The rate increases with higher thermal conductivity (a material property), larger cross-sectional area, larger temperature difference, and smaller thickness. This equation assumes steady-state conditions: the temperature at each point in the slab is constant in time, and the temperature profile is linear across the slab.

Example — Wall conduction A concrete wall ($k = 1.7 \text{ W/m}\cdot\text{K}$) is 0.20 m thick and 10 m^2 in area. Temperature difference 20°C . Rate: $P = 1.7 \times 10 \times 20 / 0.20 = 1700 \text{ W}$. This 1700 W must be continuously supplied by the heating system to maintain the indoor temperature.

30.5 Thermal Conductivity and Insulation

Thermal conductivity k characterizes how readily a material conducts heat. High- k materials (metals) conduct heat quickly; low- k materials (air, foam, fiberglass) conduct heat slowly and are used as insulators.

Material	k ($W/m\cdot K$)	Category
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Diamond	2000	Exceptional conductor
Silver	430	Metal conductor
Copper	400	Metal conductor
Aluminium	237	Metal conductor
Steel (carbon)	50	Metal conductor
Glass	0.8	Poor conductor
Concrete	1.7	Poor conductor
Water (liquid)	0.60	Poor conductor
Wood (pine)	0.12	Insulator
Fiberglass (batt)	0.040	Insulator
Air (still)	0.026	Excellent insulator
Aerogel	0.015	Extreme insulator

Table 30.1 *Thermal conductivity of common materials at approximately 25°C.*

Thermal resistance (R-value) is the engineering measure of insulating effectiveness. $R = d/k$ for a slab of thickness d and conductivity k . Higher R-value means better insulation. The total heat loss rate is $P = A \Delta T / R$. For composite walls (multiple layers), R-values add in series: $R_{\text{total}} = R_1 + R_2 + R_3 + \dots$. This is directly analogous to electrical resistance in series.

Air is an excellent insulator ($k = 0.026 \text{ W/m}\cdot\text{K}$) — but only when still. Fiberglass and foam insulation work by trapping small pockets of air and preventing it from moving (convecting). The fiberglass itself contributes little to the insulation; the trapped air does the work. Double-glazed windows create a sealed air gap between two panes to exploit this effect.

Study Tip: Good thermal insulators have low k . Good thermal conductors have high k . When a material feels cold to the touch even though it is at room temperature, it is conducting heat away from your hand rapidly — it is a good conductor. Materials that feel warm at room temperature are poor conductors (insulators), conducting less heat away from your hand.

30.6 Convection

Convection is the transfer of heat by bulk movement of a fluid (liquid or gas). In convection, warmer fluid carries thermal energy as it moves from hot regions to cooler regions. Unlike conduction, where energy is transferred between stationary molecules by collisions, convection involves the physical transport of energetic molecules through space.

The physical mechanism: when a fluid is heated, it typically becomes less dense (thermal expansion, Chapter 29). Less-dense, warmer fluid is buoyed upward by the denser, cooler fluid above it. This creates a current: warm fluid rises, carrying energy with it; cool fluid descends to replace it, gets heated, rises, and the cycle continues. This is thermal convection, and it transfers heat far more effectively than conduction alone in most fluids.

Convection is why hot air rises (hot-air balloons, chimney smoke), why the ocean circulates (thermohaline circulation), why weather systems form (differential heating of land and sea), why the Earth's mantle slowly circulates (convection cells driving tectonic plate movement), and why the Sun's outer layers show convective cells (granulation visible through telescopes).

The rate of convective heat transfer from a surface to a fluid is approximately $P = hA\Delta T$, where h is the convective heat transfer coefficient ($\text{W}/\text{m}^2\cdot\text{K}$), A is the surface area, and ΔT is the temperature difference between the surface and the far-field fluid. The value of h depends strongly on the geometry, fluid velocity, and fluid properties — it is typically measured experimentally rather than calculated from first principles.

30.7 Natural vs. Forced Convection

Natural (free) convection is driven entirely by buoyancy: the density differences that arise from temperature differences in the fluid, with no external forcing. A radiator heating a room is the classic example: the hot radiator surface warms adjacent air, which rises, cool air flows in to replace it, and a circulation pattern develops spontaneously. Natural convection is relatively gentle and slow.

Forced convection uses an external mechanism (a fan, pump, wind, or stirrer) to drive fluid motion, greatly increasing the rate of heat transfer. A cooling fan on a computer processor, a car's radiator with a fan, forced-air central heating, wind on your skin, and stirring a hot drink are all examples of forced convection. Forced convection can produce heat transfer rates 10–100 times higher than natural convection for the same temperature difference and surface area.

This is why blowing on hot soup cools it faster than waiting, why wind makes cold

weather feel colder (wind chill), and why electronics manufacturers use cooling fans rather than relying on natural convection alone. The engineering of heat exchangers (car radiators, HVAC systems, power plant condensers) is primarily the engineering of forced convection — designing geometries that maximize h for a given fluid flow rate.

Situation	Mechanism	Typical h (W/m ² ·K)
Air: natural convection	Buoyancy-driven	5–25
Air: forced convection (fan)	External flow	25–250
Water: natural convection	Buoyancy-driven	200–1000
Water: forced convection (pump)	External flow	1000–15 000
Boiling water	Phase change + convection	2500–25 000
Condensing steam	Phase change + convection	5000–100 000

Table 30.2 *Approximate convective heat transfer coefficients h for common situations.*

30.8 Radiation

Thermal radiation is the transfer of energy by electromagnetic waves emitted by any object with temperature above absolute zero. Unlike conduction and convection, radiation requires no medium — it propagates through vacuum at the speed of light. The Sun heats the Earth entirely by radiation across 150 million km of essentially empty space.

All objects simultaneously emit and absorb thermal radiation. The net heat transfer by radiation between an object and its surroundings depends on the difference between emission and absorption. An object emits radiation according to the Stefan-Boltzmann Law:

$$P_{\text{emit}} = \epsilon \sigma A T^4$$

Stefan-Boltzmann Law for emission. ϵ = emissivity (0 to 1, dimensionless); $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$ (Stefan-Boltzmann constant); A = surface area; T = absolute temperature in kelvin.

The net radiation heat transfer from an object at temperature T in surroundings at T_{surr} is:

$$P_{\text{net}} = \epsilon \sigma A (T^4 - T_{\text{surr}}^4)$$

Net radiation power. Positive means the object is emitting more than absorbing (net cooling). The T^4 dependence makes radiation increasingly dominant at high temperatures.

30.9 Blackbody Radiation (Introductory)

A blackbody is an idealized object that absorbs all electromagnetic radiation incident on it (reflecting none) and emits radiation at the maximum possible rate for its temperature. For a blackbody, the emissivity $\epsilon = 1$. Real objects have $\epsilon < 1$; the closer to 1, the closer the object's behavior is to a blackbody.

Blackbody radiation has a characteristic spectrum: the emitted radiation is spread across a range of wavelengths, with a peak wavelength that depends on temperature. As temperature increases, the peak wavelength shifts to shorter (more energetic) wavelengths. This is Wien's Displacement Law:

$$\lambda_{\text{max}} T = 2.90 \times 10^{-3} \text{ m} \cdot \text{K}$$

Wien's Displacement Law. λ_{max} = wavelength of peak emission (m); T = absolute temperature (K). Higher temperature $\rightarrow$ shorter peak wavelength.

At room temperature (300 K), $\lambda_{\text{max}} = 2.90 \times 10^{-3} / 300 = 9.7 \text{ } \mu\text{m}$: infrared radiation, invisible to the human eye. This is why thermal cameras detect people and animals in complete darkness. At the Sun's surface (5800 K), $\lambda_{\text{max}} = 2.90 \times 10^{-3} / 5800 = 500 \text{ nm}$: yellow-green visible light. The Sun's spectrum peaks in the visible precisely because biological eyes evolved to detect the Sun's most intense radiation.

Emissivity ϵ depends on both the material and its surface finish. Polished metals have very low emissivity (0.02–0.10) and are poor radiators. Matte black surfaces have high emissivity (≈ 0.95 –0.99). Space blankets (emergency foil blankets) reflect radiation and have low emissivity, dramatically reducing heat loss by radiation for a stranded person.

Surface	Emissivity ϵ	Notes
Polished silver	0.02	Excellent reflector, poor radiator
Polished aluminium	0.05	Good reflector
Polished steel	0.07	Good reflector
Oxidized steel	0.60	Much higher than polished
Human skin	0.95	Nearly a blackbody in infrared

Black paint	0.97	Near-blackbody emitter
Water (liquid)	0.95–0.96	Near-blackbody emitter
Snow	0.98	Near-blackbody emitter

Table 30.3 *Thermal emissivity of selected materials in the infrared.*

30.10 Combined Heat Transfer in Real Systems

Real-world heat transfer almost always involves all three mechanisms simultaneously. Analyzing a real system requires identifying which mechanism dominates under the given conditions and combining the effects when necessary.

A hot water radiator in a room illustrates all three: the hot water inside the radiator heats the metal casing by convection; the metal casing conducts heat to its outer surface; the outer surface heats nearby air by conduction into a thin boundary layer, which then convects away; and the outer surface also radiates infrared. Engineers must account for all three contributions when designing heating systems.

The thermal resistance analogy (parallel to electrical circuit analysis) is a powerful tool for combined systems. Each heat transfer path has a thermal resistance $R_{th} = \Delta T/P$. For paths in series (like conduction through a composite wall), R_{th} values add. For paths in parallel (like radiation and convection from a surface occurring simultaneously), conductances ($1/R_{th}$) add. This allows complex multi-layer, multi-mechanism problems to be solved systematically.

The dominance of each mechanism depends strongly on temperature. At low temperatures (room temperature and below), conduction and convection dominate for typical engineering materials. As temperature rises above a few hundred Celsius, radiation becomes increasingly significant ($P_{rad} \propto T^4$). At very high temperatures (furnaces, power plant boilers, rocket nozzles), radiation can be the dominant mechanism even in the presence of flowing gas.

30.11 Heating and Cooling Curves (Conceptual Introduction)

A heating or cooling curve tracks the temperature of a substance as it gains or loses heat over time. Understanding the shape of these curves requires combining the heat transfer mechanisms with the thermal properties of the substance (Chapter 31 gives the quantitative details).

When a cold object is placed in a warm environment: initially, the temperature difference is large, so the rate of heat transfer is high and the temperature rises quickly. As the temperature difference decreases (approaching equilibrium), the rate of transfer slows and the temperature rise decelerates. The curve is not linear — it is exponential in the purely convective/conductive case (Newton's Law of Cooling).

Newton's Law of Cooling (qualitative): the rate of cooling of an object is approximately proportional to the temperature difference between the object and its surroundings. This produces an exponential approach to equilibrium: $\Delta T(t) = \Delta T_0 e^{(-t/\tau)}$, where τ is the thermal time constant of the system. Large thermal mass (heavy, high-specific-heat objects) or low heat transfer coefficient (good insulation) gives large τ and slow equilibration.

When a substance also undergoes a phase change during heating (melting or boiling), a horizontal plateau appears on the temperature-time graph. At the phase change temperature, all added energy goes into changing the state of the material rather than raising its temperature. This plateau is a dramatic signature of latent heat, which is the subject of Chapter 31.

30.12 Engineering and Everyday Applications of Heat Transfer

Heat transfer engineering is one of the most practically important branches of applied physics. It underpins the design of buildings, electronic devices, power plants, vehicles, food processing systems, medical equipment, and spacecraft.

Building insulation and energy efficiency

Building energy codes specify minimum R-values for walls, roofs, and windows. Double and triple glazing with argon or krypton fill (low-k gases), low-emissivity coatings on glass surfaces (reducing radiation), and insulated concrete forms all reduce conduction and radiation losses. Thermal bridges — high-conductivity paths through an otherwise insulating structure (a steel stud in a foam wall) — are a major source of heat loss and must be minimized.

Electronic cooling

Microprocessors dissipate 50–300 W in a chip smaller than a postage stamp, producing heat fluxes that would destroy the chip without active cooling. Copper or aluminium heatsinks with finned surfaces increase area for conduction and convection. Heat pipes use phase-change heat transfer (evaporation and condensation) to move heat from the processor to a remote radiator with near-zero thermal resistance. Liquid cooling loops circulate water or coolant directly over the chip for even higher heat transfer.

Thermos (vacuum flask)

A thermos bottle eliminates all three heat transfer mechanisms. The vacuum between inner and outer glass walls eliminates conduction and convection (no medium for either). The silvered surfaces have near-zero emissivity, minimizing radiation. Result: a beverage can stay hot or cold for hours with minimal heat exchange with the environment.

Climate and atmospheric science

The greenhouse effect is a radiation heat transfer phenomenon. Short-wavelength solar radiation passes through the atmosphere and heats the Earth's surface. The surface emits long-wavelength (infrared) radiation, which is partially absorbed and re-emitted by greenhouse gases (CO_2 , H_2O , CH_4) in the atmosphere, reducing the net heat loss to space. This keeps Earth's average temperature about 33°C warmer than it would be without an atmosphere, but increasing greenhouse gas concentrations are enhancing this effect beyond the natural level.

Historical Note: Joseph Fourier (1768–1830) first recognized that the atmosphere acts as an insulating layer — the first description of what we now call the greenhouse effect. Svante Arrhenius (1859–1927) calculated in 1896 that doubling atmospheric CO_2 would raise Earth's average temperature by $5\text{--}6^\circ\text{C}$. The modern consensus of $\approx 3^\circ\text{C}$ per doubling of CO_2 is remarkably close to Arrhenius's estimate, made without computers.

30.13 Common Mistakes in Heat Transfer Analysis

- Confusing which mechanism applies. Conduction occurs in solids and still fluids through molecular contact. Convection requires bulk fluid motion. Radiation requires no medium. A common error is treating vacuum gaps as conductive (e.g., assuming heat crosses a vacuum by conduction).
- Forgetting that radiation depends on T^4 , not ΔT . The Stefan-Boltzmann Law gives emission proportional to T^4 . Net radiation transfer is proportional to $T^4 - T_{\text{surr}}^4$, not to $\Delta T = T - T_{\text{surr}}$. At small ΔT , these are approximately proportional; at large ΔT , the T^4 dependence becomes crucial.
- Using Celsius instead of Kelvin in radiation calculations. The T^4 term in the Stefan-Boltzmann Law requires absolute temperature in kelvins. Using Celsius gives completely wrong results. This is the most common computational error in radiation problems.
- Omitting thermal resistance of air gaps in composite wall problems. An air gap between two surfaces has its own R-value (due to the convection resistance on

each surface and the conduction/radiation across the gap). Ignoring it significantly overestimates the heat transfer rate.

- Treating the Fourier conduction equation as applying to non-steady-state situations. $P = kA\Delta T/d$ assumes the temperature at every point in the slab is constant in time. If temperature is changing, the full time-dependent diffusion equation must be used.
- Assuming high emissivity surfaces are good conductors. Emissivity governs radiation; conductivity governs conduction. Black paint has very high emissivity but very low conductivity. Polished metal has very low emissivity but very high conductivity. These are completely independent material properties.

Common Mistake: In all radiation calculations, convert temperatures to Kelvin before applying $P = \epsilon\sigma AT^4$ or $P_{\text{net}} = \epsilon\sigma A(T^4 - T_{\text{surr}}^4)$. Celsius temperatures in these equations give results off by orders of magnitude.

Chapter Summary

Heat transfers from higher to lower temperature by three mechanisms: conduction, convection, and radiation. All three are simultaneously present in most real systems.

- Conduction: $P = kA\Delta T/d$. Operates through molecular contact. Metals conduct well (free electrons). Insulators trap still air.
- Convection: $P = hA\Delta T$. Bulk fluid motion carries energy. Natural convection driven by buoyancy; forced convection by external flow. Forced convection is far more effective.
- Radiation: $P = \epsilon\sigma AT^4$; net $P = \epsilon\sigma A(T^4 - T_{\text{surr}}^4)$. Requires no medium. All objects emit and absorb radiation. T must be in kelvins.
- Blackbody: ideal emitter with $\epsilon = 1$. Wien's Law: $\lambda_{\text{max}} T = 2.90 \times 10^{-3} \text{ m}\cdot\text{K}$. Higher $T \rightarrow$ shorter peak wavelength.
- Thermal resistance $R_{\text{th}} = \Delta T/P = d/(kA)$ for conduction slabs; adds in series for composite walls.
- Radiation dominates at high temperatures (T^4 dependence); conduction and convection dominate at moderate temperatures.
- Applications: building insulation (minimize conduction/radiation), electronic cooling (maximize convection), thermos (eliminate all three), greenhouse effect (radiation trapping by atmosphere).

Key Terms

heat transfer: The movement of thermal energy from a region of higher temperature to a region of lower temperature by conduction, convection, or radiation.

conduction: Heat transfer through a material by molecular vibration and electron transport without bulk motion of the material; rate $P = kA\Delta T/d$.

thermal conductivity (k): Material property describing the rate of conductive heat flow per unit area per unit temperature gradient; units $\text{W/m}\cdot\text{K}$. High k : good conductor; low k : good insulator.

thermal resistance (R-value): $R = d/k$ for a material slab; measures insulating effectiveness; higher R-value means better insulation; adds in series for composite walls.

convection: Heat transfer by bulk motion of a fluid (gas or liquid) carrying thermal energy from hot to cool regions.

natural convection: Convection driven by buoyancy forces arising from temperature-induced density differences in the fluid; no external forcing.

forced convection: Convection enhanced by external mechanical forcing (fans, pumps, wind); produces much higher heat transfer rates than natural convection.

convective heat transfer coefficient (h): Empirical constant relating convective heat transfer rate to temperature difference and surface area: $P = hA\Delta T$; units $\text{W/m}^2\cdot\text{K}$.

radiation (thermal): Heat transfer by electromagnetic waves emitted by all objects above absolute zero; requires no medium; rate $P = \epsilon\sigma AT^4$.

Stefan-Boltzmann Law: $P = \epsilon\sigma AT^4$; total power emitted by a surface at absolute temperature T ; $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4$.

emissivity (ϵ): Dimensionless property (0 to 1) measuring how effectively a surface emits (and absorbs) thermal radiation compared to a perfect blackbody.

blackbody: An idealized object that absorbs all incident radiation and emits at the maximum possible rate for its temperature; $\epsilon = 1$.

Wien's Displacement Law: $\lambda_{\text{max}} T = 2.90 \times 10^{-3} \text{ m}\cdot\text{K}$; the peak wavelength of blackbody emission shifts to shorter wavelengths at higher temperatures.

Newton's Law of Cooling: The rate of heat loss of an object is proportional to the temperature difference between the object and its surroundings; leads to exponential temperature decay.

greenhouse effect: The trapping of outgoing infrared radiation by atmospheric gases (CO_2 , H_2O , CH_4), warming Earth's surface above what solar radiation alone would produce.

thermal time constant (τ): The characteristic time for a cooling/heating process to

reduce the temperature difference by a factor of $1/e$; $\tau = (\text{thermal mass})/(\text{heat transfer coefficient} \times \text{area})$.

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. A metal spoon feels colder than a wooden spoon when both are at room temperature. Explain this using the concept of thermal conductivity. Do the spoons actually have different temperatures?
 2. Explain why a thermos bottle keeps drinks hot (or cold). Identify which heat transfer mechanism each design feature addresses.
 3. The Sun heats the Earth across 150 million km of vacuum. Which heat transfer mechanism is responsible? Why cannot conduction or convection account for this?
 4. Explain why forced convection (a fan) cools a hot surface much more effectively than natural convection. What physical quantity does the fan increase?
 5. A room is heated by a cast-iron radiator. Identify the role of conduction, convection, and radiation in distributing heat from the radiator to the room.
 6. Why must temperature be expressed in kelvins when using the Stefan-Boltzmann Law? Give a numerical example showing how large the error is when Celsius is used instead of Kelvin for an object at 27°C .
 7. Explain the greenhouse effect using the concept of thermal radiation. Why does increasing atmospheric CO_2 concentration raise Earth's average surface temperature?
 8. A polished aluminium surface and a matte black surface are both at 200°C . Which emits more thermal radiation? Which is a better heat sink for electronic cooling if cooled by forced-air convection? Explain.
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Heat Transfer Identification Exercises

For each situation, identify the primary heat transfer mechanism(s) and explain briefly.

9. Heat moves from a hot stove burner through a copper pot to the water inside.

10. The Sun warms the surface of the Moon.
 11. Air above a heated parking lot rises and is replaced by cooler air.
 12. A person feels warm near a campfire even without touching it.
 13. Ice forms on a lake surface on a cold night (the lake loses heat to the air).
 14. A computer processor is cooled by a liquid cooling loop with a pump.
 15. A firefighter in proximity to flames is protected by a reflective suit.
 16. The Earth's mantle slowly circulates, driving plate tectonics.
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Conduction Problems

17. A glass window ($k = 0.80 \text{ W/m}\cdot\text{K}$) is $1.2 \text{ m} \times 0.8 \text{ m}$ and 6 mm thick. The inside surface is at 20°C and the outside at -5°C . Find the rate of heat loss through the window.
 18. A composite wall consists of an 8 cm layer of brick ($k = 0.72 \text{ W/m}\cdot\text{K}$) and a 5 cm layer of fiberglass ($k = 0.040 \text{ W/m}\cdot\text{K}$). The inner surface is at 22°C and the outer surface at -20°C . Area = 15 m^2 . Find (a) the thermal resistance of each layer, (b) the total thermal resistance, and (c) the heat loss rate.
 19. A copper rod ($k = 400 \text{ W/m}\cdot\text{K}$, diameter 2.0 cm , length 50 cm) connects a heat source at 100°C to a heat sink at 20°C . Find the heat conduction rate.
 20. Adding 10 cm of foam insulation ($k = 0.030 \text{ W/m}\cdot\text{K}$) to the wall in Problem 2 reduces heat loss by what factor? (Add the foam as a third layer in series.)
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Convection and Radiation Analysis

21. A person's skin surface area is 1.8 m^2 and skin temperature is 34°C . The surrounding air is at 20°C . With a convective coefficient $h = 8 \text{ W/m}^2\cdot\text{K}$, what is the convective heat loss rate?
22. Calculate the radiation emitted by a human body (skin temperature 307 K , $\varepsilon = 0.95$, area 1.8 m^2) in a room at 293 K . What is the net radiation heat loss?

23. The Sun's surface temperature is 5800 K. Using Wien's Displacement Law, find the peak wavelength of its radiation. Is this in the infrared, visible, or ultraviolet range?
24. A blackbody at 3000 K (hot tungsten filament) radiates with area $5.0 \times 10^{-5} \text{ m}^2$. Find its power output. By what factor would the power increase if the temperature were doubled to 6000 K?

Applied Thermal Systems Problems

25. A thermos has a vacuum gap between two silvered glass walls. Estimate the radiation heat loss rate if the inner surface (holding hot coffee at $80^\circ\text{C} = 353 \text{ K}$) has $\epsilon = 0.02$, area 0.030 m^2 , and the outer wall is at $20^\circ\text{C} = 293 \text{ K}$. (Use net radiation formula.)
26. A building wall (area 80 m^2) consists of three layers: interior plasterboard ($k = 0.20 \text{ W/m}\cdot\text{K}$, $d = 12 \text{ mm}$), fiberglass insulation ($k = 0.040 \text{ W/m}\cdot\text{K}$, $d = 90 \text{ mm}$), and exterior brick ($k = 0.72 \text{ W/m}\cdot\text{K}$, $d = 100 \text{ mm}$). Inside 20°C , outside 5°C . Find the total heat loss rate.
27. A microprocessor dissipates 95 W. Its heatsink has area 0.008 m^2 and is cooled by forced air with $h = 180 \text{ W/m}^2\cdot\text{K}$. What is the temperature difference between the heatsink surface and the cooling air? (Hint: $P = hA\Delta T$.)
28. Earth receives solar irradiance of 1361 W/m^2 at the top of the atmosphere. Earth's albedo (fraction reflected) is 0.30. Earth radiates as a near-blackbody ($\epsilon \approx 1$). Find Earth's equilibrium temperature (the temperature at which radiated power equals absorbed solar power). Compare to Earth's actual average surface temperature of 288 K. What accounts for the difference?

Chapter 30 Checkpoint

Before moving on to Chapter 31, confirm you can do each of the following.

I can...	See Section
Distinguish heat transfer from heat and thermal energy	30.1–30.2
Explain why heat flows from higher to lower temperature	30.3

State Fourier's Law and calculate conduction rate $P = kA\Delta T/d$	30.4
Compare thermal conductivities of metals, non-metals, and insulators	30.5
Explain convection and calculate rate using $P = hA\Delta T$	30.6
Distinguish natural from forced convection; state why forced is more effective	30.7
Apply Stefan-Boltzmann Law: $P = \epsilon\sigma AT^4$ (in kelvins)	30.8
Apply Wien's Displacement Law to find peak radiation wavelength	30.9
Analyze composite walls using thermal resistances in series	30.10
Describe how all three mechanisms combine in real systems	30.10
Explain the greenhouse effect as a radiation heat transfer phenomenon	30.12
Identify and avoid common errors in heat transfer calculations	30.13

Continue to Chapter 31: Specific Heat and Phase Changes →